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GATE ECE · ENGINEERING MATHEMATICS · CHAPTER-WISE PYQS

Transform Theory

GATE ECE Engineering Mathematics · Previous Year Questions with Answer Key

18	15	1995 – 2015	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	13 questions
Marks 2	5 questions
Answer key	at the end

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Marks 1 13 questions

Q1. GATE ECE 2015 Set 3

Consider the function $g(t) = e^{-t} \sin(2\pi t) u(t)$,where $u(t)$ is the unit step function. The area under $g(t)$ is _____.

Open solution

Q2. GATE ECE 2014 Set 4

The unilateral Laplace transform of $f(t)$ is $\frac{1}{s^2+s+1}$. Which one of the following is the unilateral Laplace transform of $g(t) = t.f(t)$?

(A) $\frac{-s}{(s^2+s+1)^2}$

(B) $\frac{-(2s+1)}{(s^2+s+1)^2}$

(C) $\frac{s}{(s^2+s+1)^2}$

(D) $\frac{2s+1}{(s^2+s+1)^2}$

Open solution

Q3. GATE ECE 2012

If $x[N] = (1/3)^{|n|} - (1/2)^n u[n]$, then the region of convergence (ROC) of its Z-transform in the Z-plane will be

(A) $\frac{1}{3} < |z| < 3$

(B) $\frac{1}{3} < |z| < \frac{1}{2}$

(C) $\frac{1}{2} < |z| < 3$

(D) $\frac{1}{3} < |z|$

Open solution

Q4. GATE ECE 2012

The unilateral Laplace transform of $f(t)$ is $\frac{1}{s^2+s+1}$. The unilateral Laplace transform of $t\ f(t)$ is

(A) $-\frac{s}{(s^2+s+1)^2}$

(B) $-\frac{2s+1}{(s^2+s+1)^2}$

(C) $\frac{s}{(s^2+s+1)^2}$

(D) $\frac{2s+1}{(s^2+s+1)^2}$

Open solution

Q5. GATE ECE 2009

Given that $F(s)$ is the one-sided Laplace transform of $f(t)$, the Laplace transform of $\int_0^t f(\tau)d\tau$ is

(A) $s\ F(s) - f(0)$

(B) $\frac{1}{s}F(s)$

(C) $\int_0^s f(\tau)d\tau$

(D) $\frac{1}{s}[F(s) - f(0)]$

Open solution

Q6. GATE ECE 2006

Consider the function $f(t)$ having laplace transform $F(s) = \frac{\omega_0}{s^2+\omega_0^2}$, $\mathrm{Re}(s) > 0$. The final value of $f(t)$ would be _____.

(A) 0

(B) 1

(C) $-1 \le f(\infty) \le 1$

(D) ∞

Open solution

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Q7. GATE ECE 2005

In what range should $Re(s)$ remain so that the laplace transform of the function $e^{(a+2)t+5}$ exists?

(A) $\mathrm{Re}\,(s) > a+2$ (B) $\mathrm{Re}\,(s) > a+7$

(C) $\mathrm{Re}\,(s) < 2$ (D) $\mathrm{Re}\,(s) > a+5$

Open solution

Q8. GATE ECE 2003

The laplace transform of $i(t)$ is given by $I(s) = \frac{2}{s(1+s)}$ As $t \to \infty$, the value of $i(t)$ tends to _____.

(A) 0 (B) 1 (C) 2 (D) ∞

Open solution

Q9. GATE ECE 1999

If $L\{f(t)\} = F(s)$ then $L\{f(t-T)\}$ is equal to

(A) $e^{s\,T}F(s)$ (B) $e^{-s\,T}F(s)$

(C) $\frac{F(s)}{1-e^{s\,T}}$ (D) $\frac{F(s)}{1-e^{-s\,T}}$

Open solution

Q10. GATE ECE 1998

If $L\ \{f(t)\} = \frac{w}{s^2+w^2}$ then the value of $\underset{t\to\infty}{Lim} f(t) =$ _____.

(A) can not be determined (B) zero

(C) unity (D) infinite

Open solution

Q11. GATE ECE 1997

The laplace transform of $e^{\alpha t}\ \cos\ \alpha\, t$ is equal to _____.

(A) $\frac{s-\alpha}{(s-\alpha)^2+\alpha^2}$ (B) $\frac{s+\alpha}{(s+\alpha)^2+\alpha^2}$

(C) $\frac{1}{(s-\alpha)^2}$ (D) None

Open solution

Q12. GATE ECE 1996

The inverse laplace transform of the function $\frac{s+5}{(s+1)(s+3)}$ is _____.

(A) $2\,e^{-t} - e^{-3t}$ (B) $2\,e^{-t} + e^{-3t}$ (C) $e^{-t} - 2\,e^{-3t}$ (D) $e^{-t} + 2\,e^{-3t}$

Open solution

Q13. GATE ECE 1995

If $L\{f(t)\} = \frac{2(s+1)}{s^2+2s+5}$ then $f(0^+)$ and $f(\propto)$ are given by _____.

(A) $0, 2$ respectively (B) $2, 0$ respectively

(C) $0, 1$ respectively (D) $\frac{2}{5}, 0$ respectively

Open solution

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Marks 2 5 questions

Q14. GATE ECE 2015 Set 2

Open solution

The bilateral Laplace transform of a function $f(t) = \begin{cases} 1 & if\ a \le t \le b \\ 0 & otherwise \end{cases}$ is

(A) $\frac{a-b}{s}$

(B) $\frac{e^s(a-b)}{s}$

(C) $\frac{e^{-as}-e^{-bs}}{s}$

(D) $\frac{e^{s(a-b)}}{s}$

Q15. GATE ECE 2014 Set 1

Open solution

A system is described by the following differential equation, where $u(t)$ is the input to the system and $y(t)$ is the output of the system.

$$\dot{y}(t) + 5y(t) = u(t)$$

When $y(0) = 1$ and $u(t)$ is a unit step function, $y(t)$ is

(A) $0.2 + 0.8e^{-5t}$ (B) $0.2 - 0.2e^{-5t}$ (C) $0.8 + 0.2e^{-5t}$ (D) $0.8 - 0.8e^{-5t}$

Q16. GATE ECE 2012

Open solution

Consider the differential equation
$\frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + y(t) = \delta(t)$
with $y(t)|_{t=0} = -2$ and $\frac{dy}{dt}|_{t=0} = 0$.
The numerical value of $\frac{dy}{dt}|_{t=0}$. is

(A) -2 (B) -1 (C) 0 (D) 1

Q17. GATE ECE 2005

Open solution

The Dirac delta Function $\delta(t)$ is defined as

(A) $\delta(t) = \begin{cases} 1 & t = 0 \\ 0\ , & otherwise \end{cases}$

(B) $\delta(t) = \begin{cases} \infty & t = 0 \\ 0\ , & otherwise \end{cases}$

(C) $\delta(t) = \begin{cases} 1 & t = 0 \\ 0\ , & otherwise \end{cases}$ and $\int_{-\infty}^{\infty} \delta(t)\,dt = 1$

(D) $\delta(t) = \begin{cases} \infty & t = 0 \\ 0\ , & otherwise \end{cases}$ and $\int_{-\infty}^{\infty} \delta(t)\,dt = 1$

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Q18. GATE ECE 2000

Open solution

If $L\{f(t)\} = \frac{s+2}{s^2+1}$, $L\{g(t)\} = \frac{s^2+1}{(s+3)(s+2)}$,
$h(t) = \int_0^t f(T)g(t-T)dT$
then $L\{h(t)\}$ is _____.

(A) $\frac{s^2+1}{s+3}$

(B) $\frac{1}{s+3}$

(C) $\frac{s^2+1}{(s+3)(s+2)} + \frac{s+2}{s^2+1}$

(D) None

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Answer Key

Marks 1

1. **0.155**	2. **D**	3. **C**	4. **D**	5. **B**
6. **A**	7. **A**	8. **C**	9. **B**	10. **A**
11. **A**	12. **A**	13. **B**		

Marks 2

14. **C**	15. **A**	16. **D**	17. **D**	18. **B**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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